

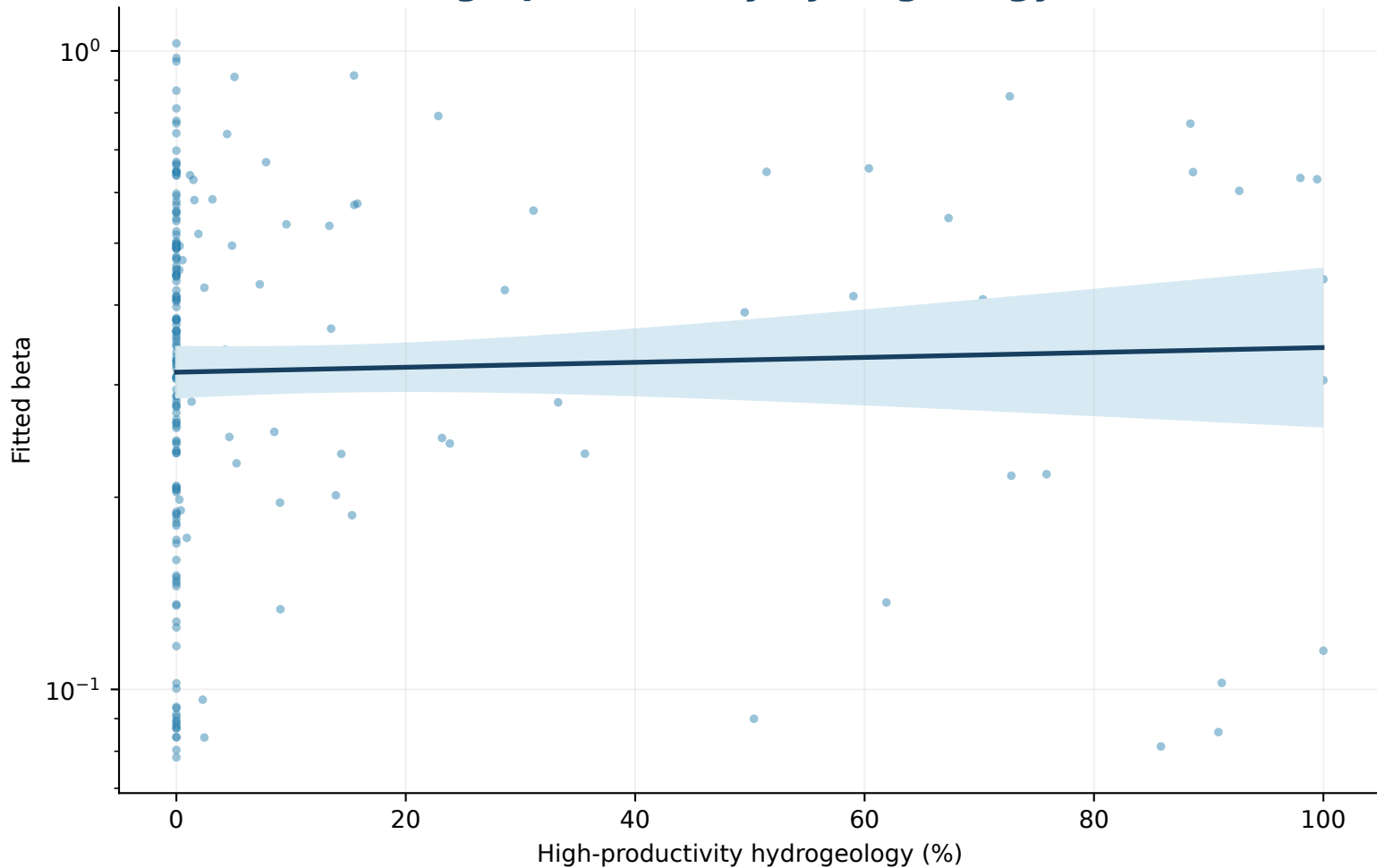
Fitted beta one-variable analysis

Target: **long-duration tail shape**. Response transformation: $\log(\text{fitted_beta})$; all eligible values are strictly positive.

Each predictor is fitted separately using OLS with HC3 robust uncertainty. Effects are exact percentage differences for frozen meaningful increments. Raw p-values and Benjamini-Hochberg FDR q-values are both reported.

Primary sample: strict, n range 212-227. Annual NIMROD indices are an independent rainfall FDR family. All associations are unadjusted and non-causal.

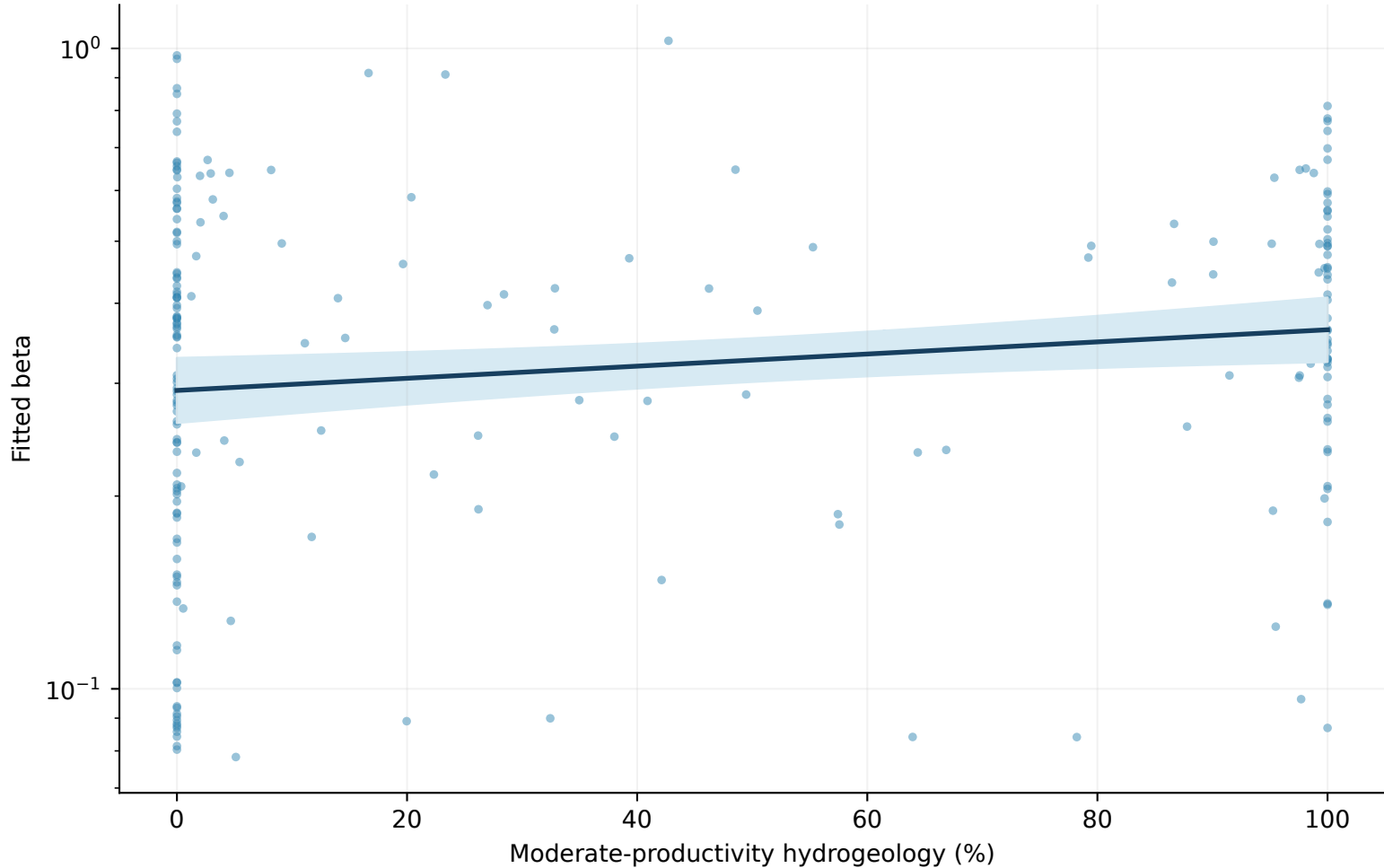
Fitted beta vs High-productivity hydrogeology



n=227 | effect +0.9% per 10 percentage points | 95% CI [-2.2, +4.1]% | p=0.576 | q=0.576 | R²=0.002

A 10 percentage points increase in High-productivity hydrogeology is associated with an estimated 0.9% higher fitted beta; unadjusted and non-c

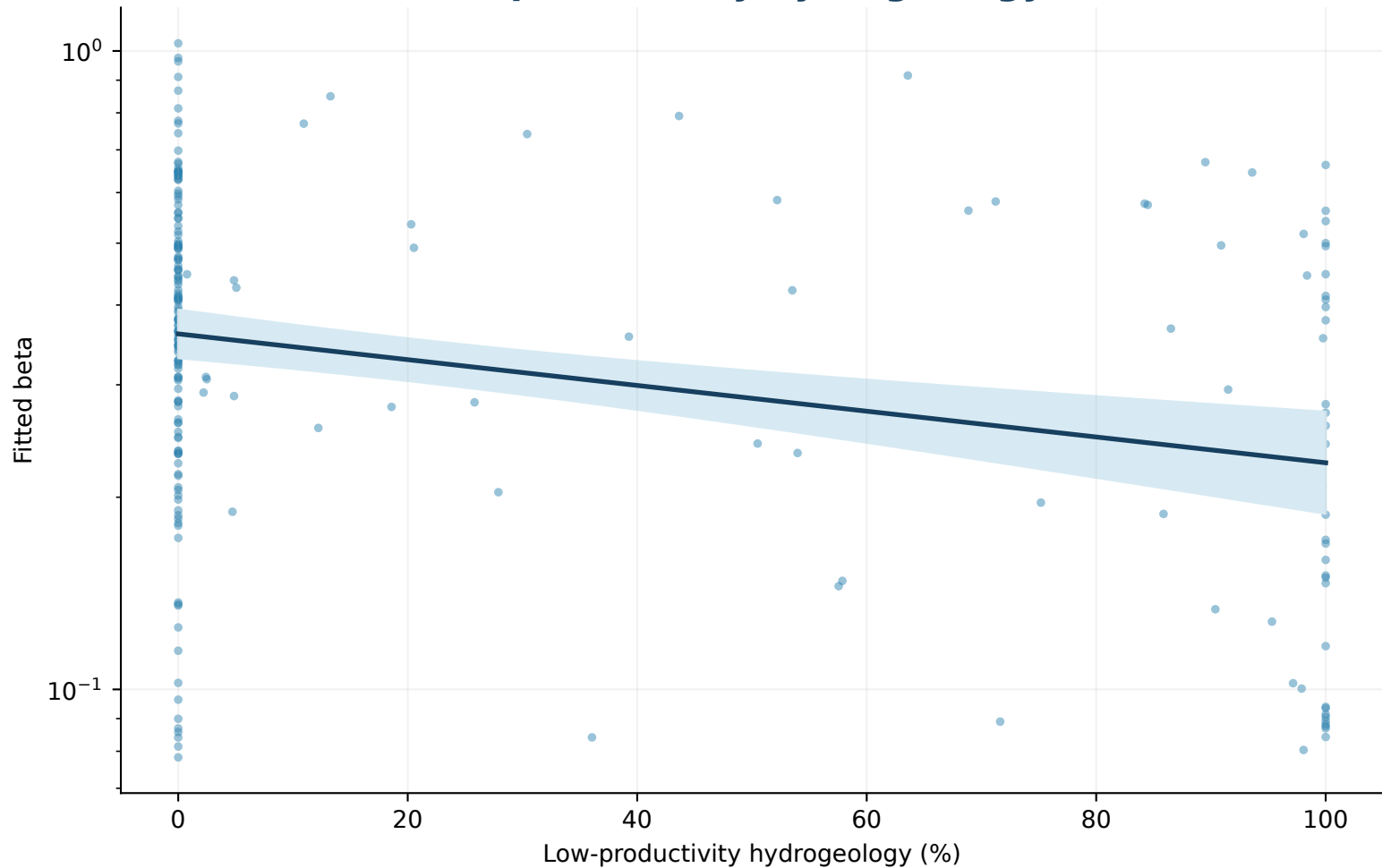
Fitted beta vs Moderate-productivity hydrogeology



n=227 | effect +2.2% per 10 percentage points | 95% CI [+0.5, +4.0]% | p=0.0128 | q=0.0207 | R2=0.024

A 10 percentage points increase in Moderate-productivity hydrogeology is associated with an estimated 2.2% higher fitted beta; unadjusted and

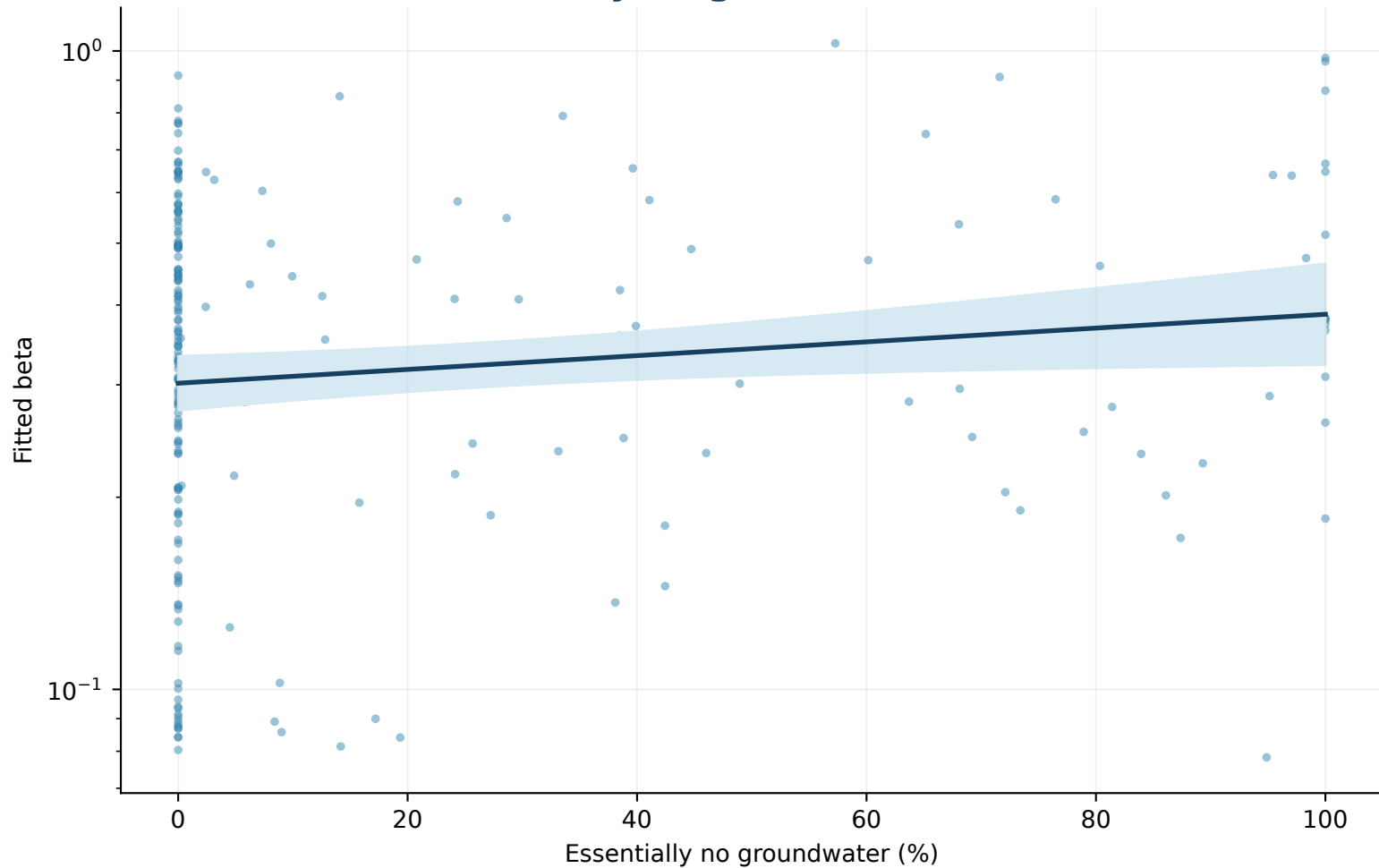
Fitted beta vs Low-productivity hydrogeology



n=227 | effect -4.5% per 10 percentage points | 95% CI [-6.5, -2.5]% | p=1.18e-05 | q=5.12e-05 | R2=0.094

A 10 percentage points increase in Low-productivity hydrogeology is associated with an estimated 4.5% lower fitted beta; unadjusted and non-ca

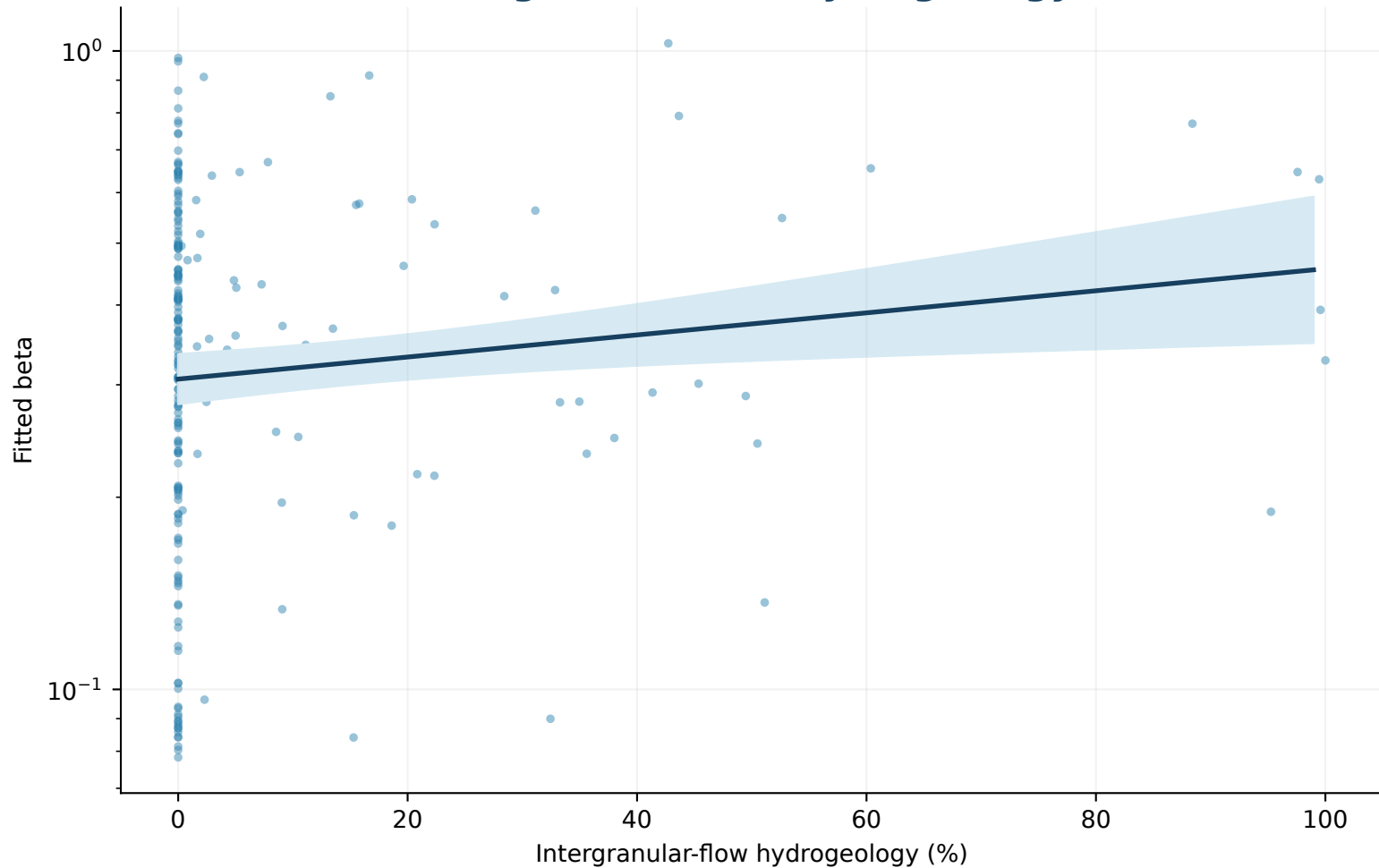
Fitted beta vs Essentially no groundwater



n=227 | effect +2.5% per 10 percentage points | 95% CI [+0.3, +4.8]% | p=0.026 | q=0.0338 | R²=0.018

A 10 percentage points increase in Essentially no groundwater is associated with an estimated 2.5% higher fitted beta; unadjusted and non-causal

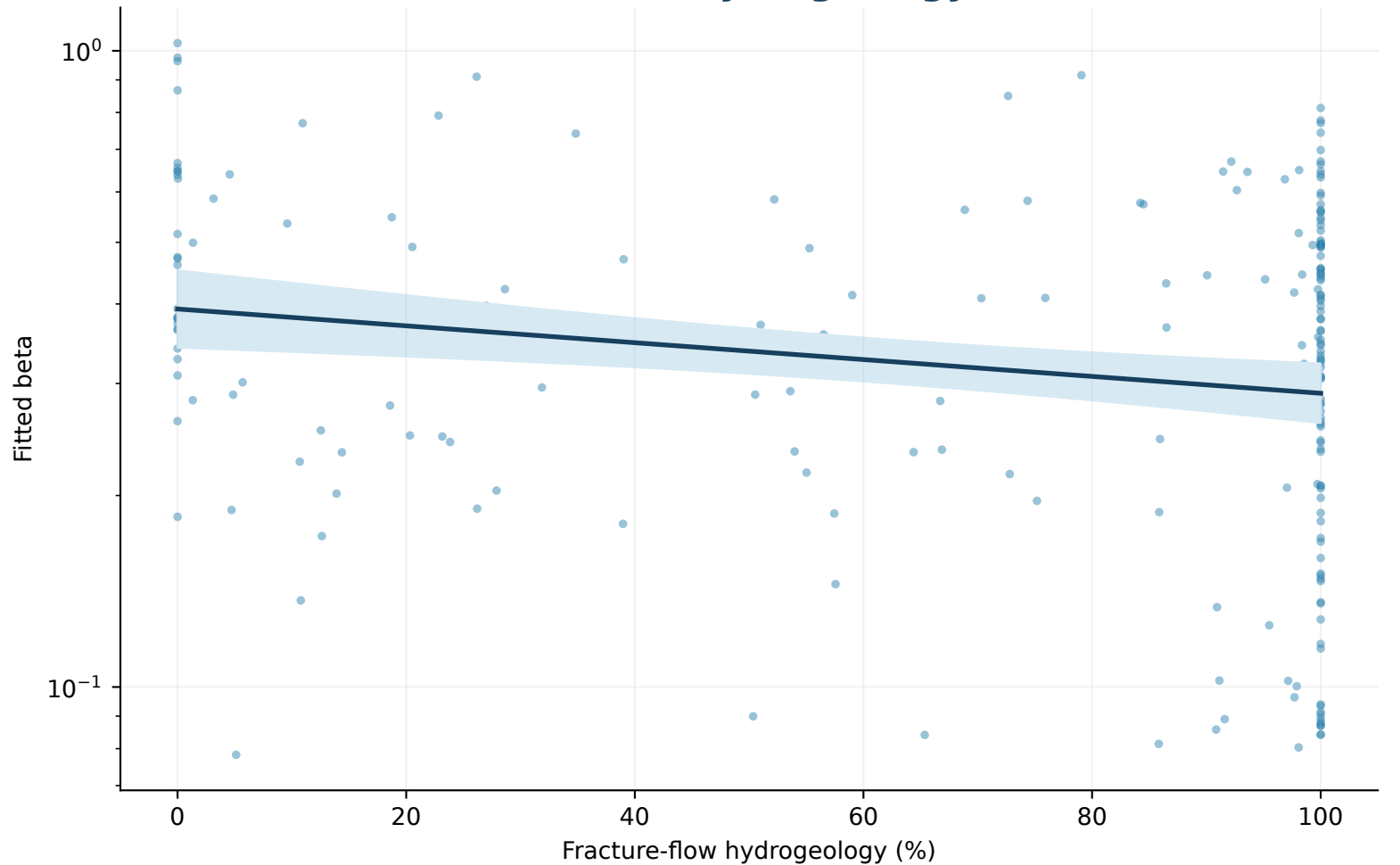
Fitted beta vs Intergranular-flow hydrogeology



n=227 | effect +4.1% per 10 percentage points | 95% CI [+1.0, +7.2]% | p=0.00834 | q=0.0155 | R²=0.019

A 10 percentage points increase in Intergranular-flow hydrogeology is associated with an estimated 4.1% higher fitted beta; unadjusted and non-

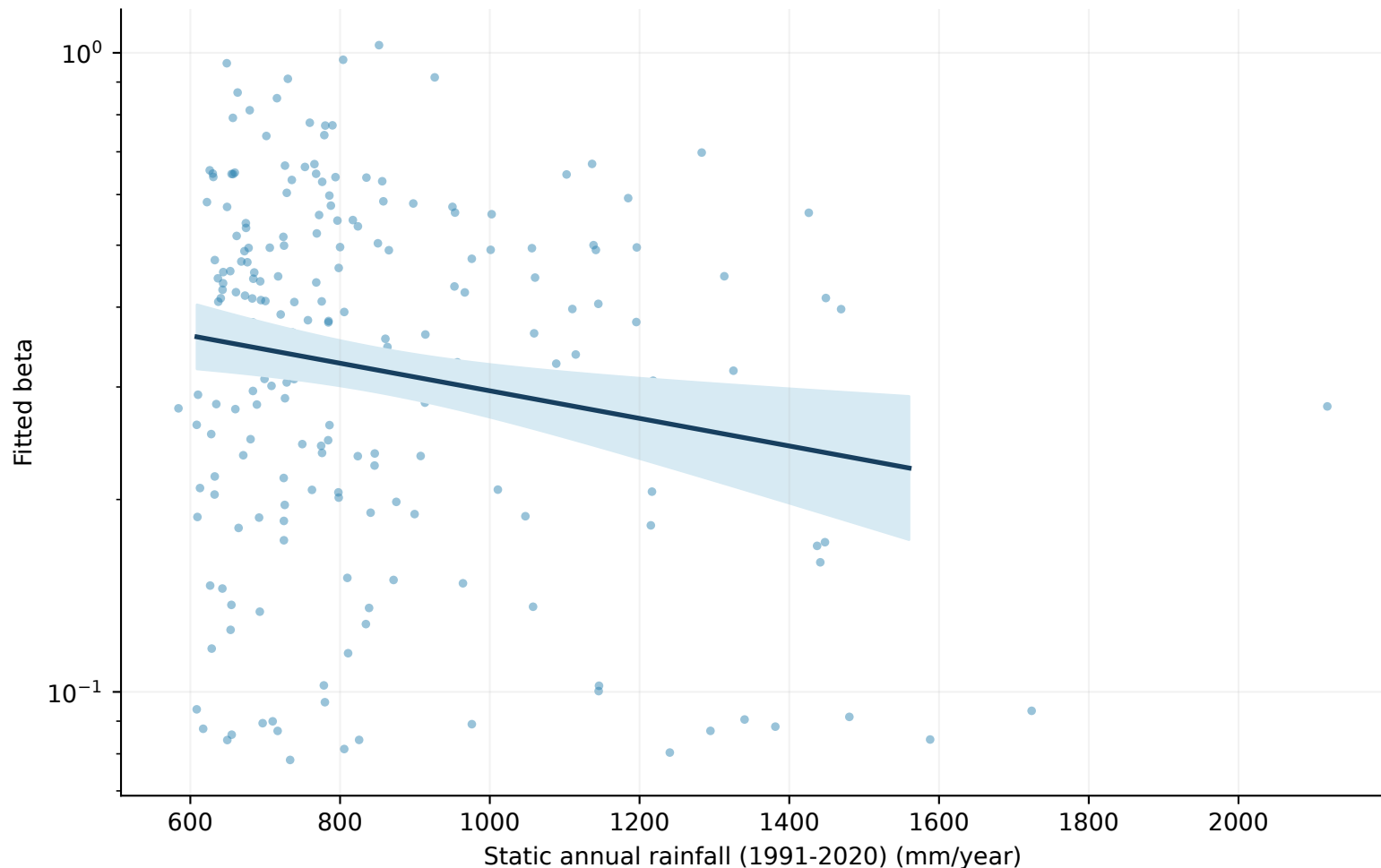
Fitted beta vs Fracture-flow hydrogeology



n=227 | effect -3.0% per 10 percentage points | 95% CI [-4.8, -1.2]% | p=0.00144 | q=0.00468 | R2=0.037

A 10 percentage points increase in Fracture-flow hydrogeology is associated with an estimated 3.0% lower fitted beta; unadjusted and non-causal

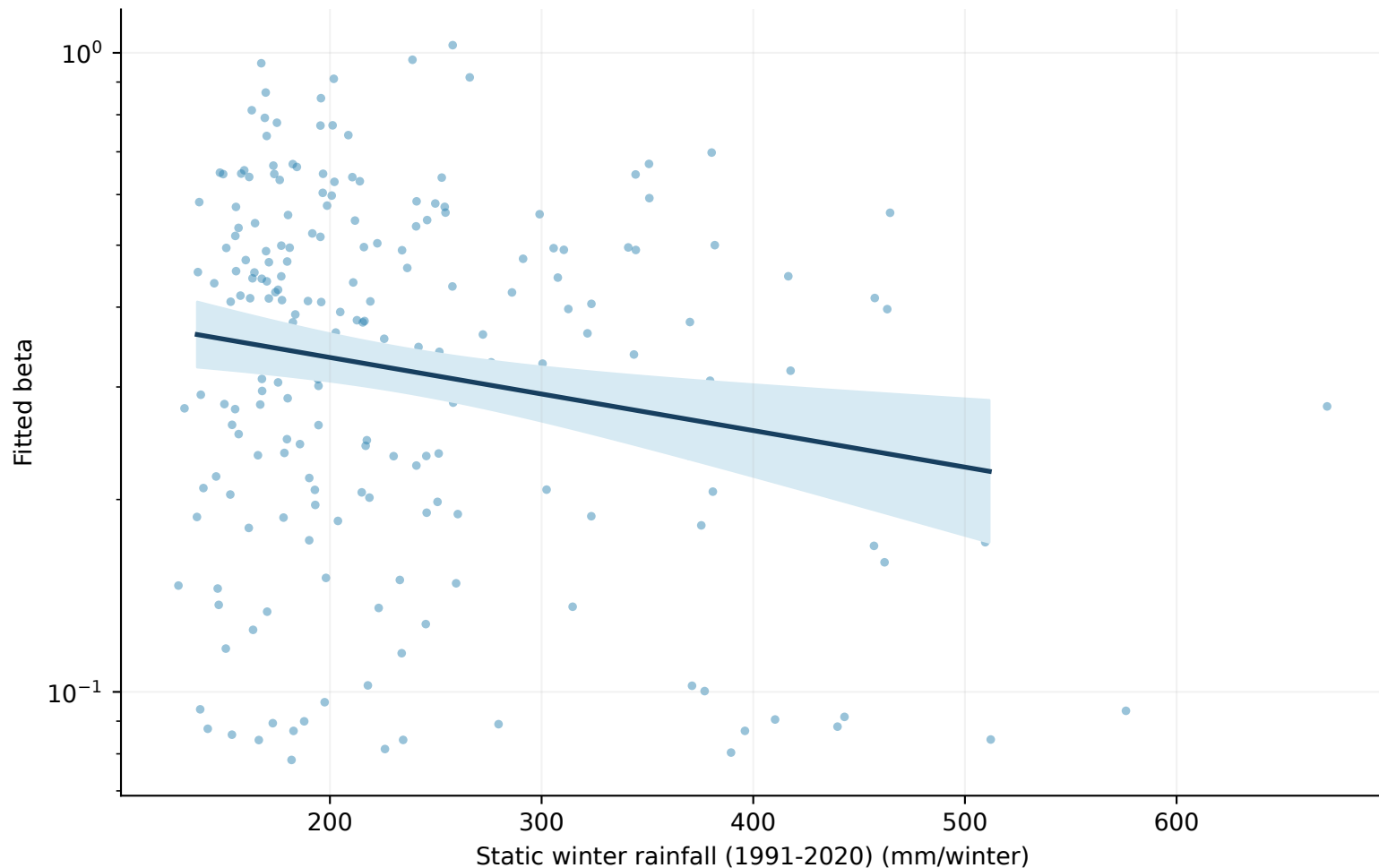
Fitted beta vs Static annual rainfall (1991-2020)



n=227 | effect -4.8% per 100 mm/year | 95% CI [-8.1, -1.5]% | p=0.00494 | q=0.0107 | R2=0.038

A 100 mm/year increase in Static annual rainfall (1991-2020) is associated with an estimated 4.8% lower fitted beta; unadjusted and non-causal.

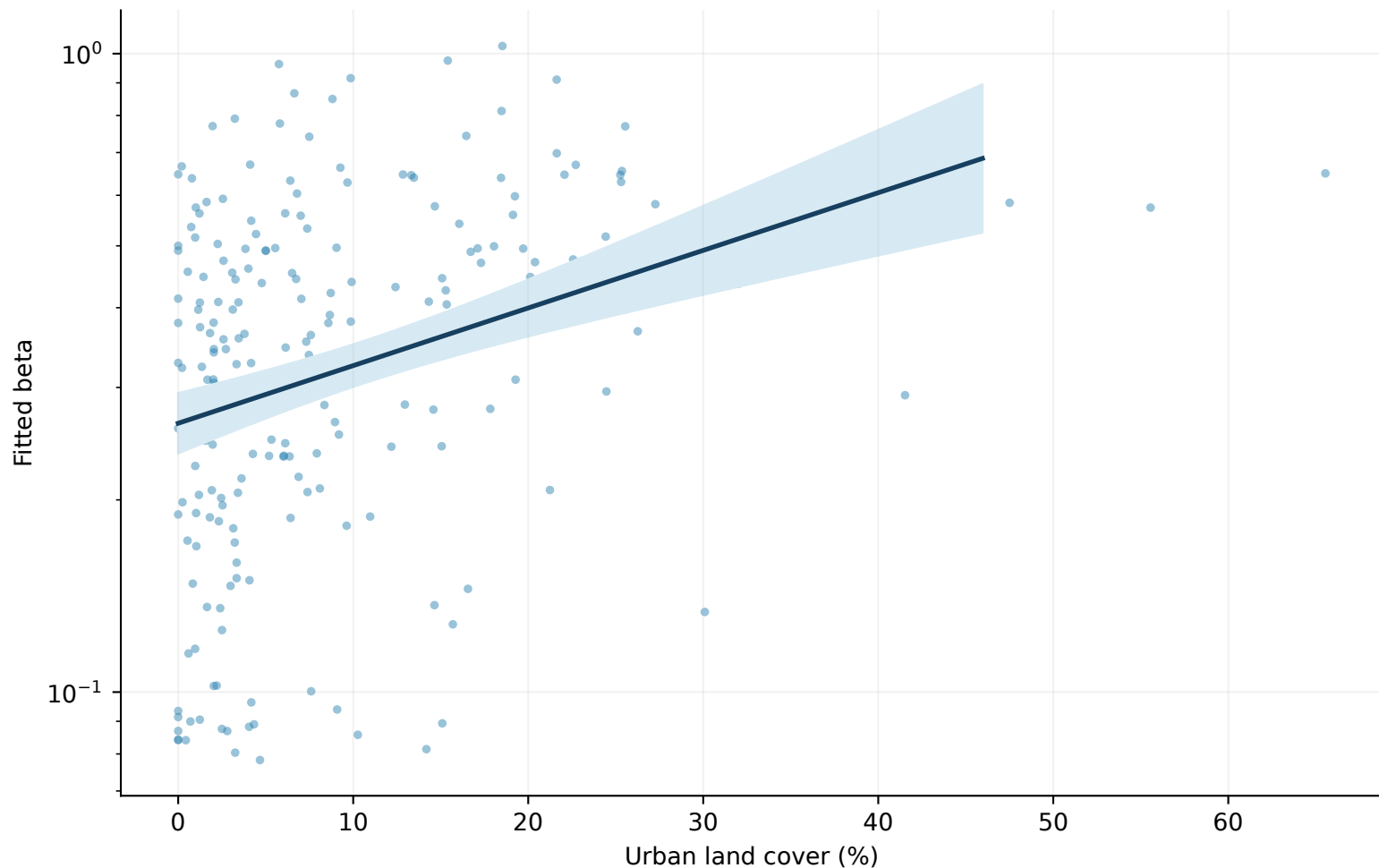
Fitted beta vs Static winter rainfall (1991-2020)



n=227 | effect -12.3% per 100 mm/winter | 95% CI [-19.7, -4.3]% | p=0.00342 | q=0.00888 | R²=0.040

A 100 mm/winter increase in Static winter rainfall (1991-2020) is associated with an estimated 12.3% lower fitted beta; unadjusted and non-causal

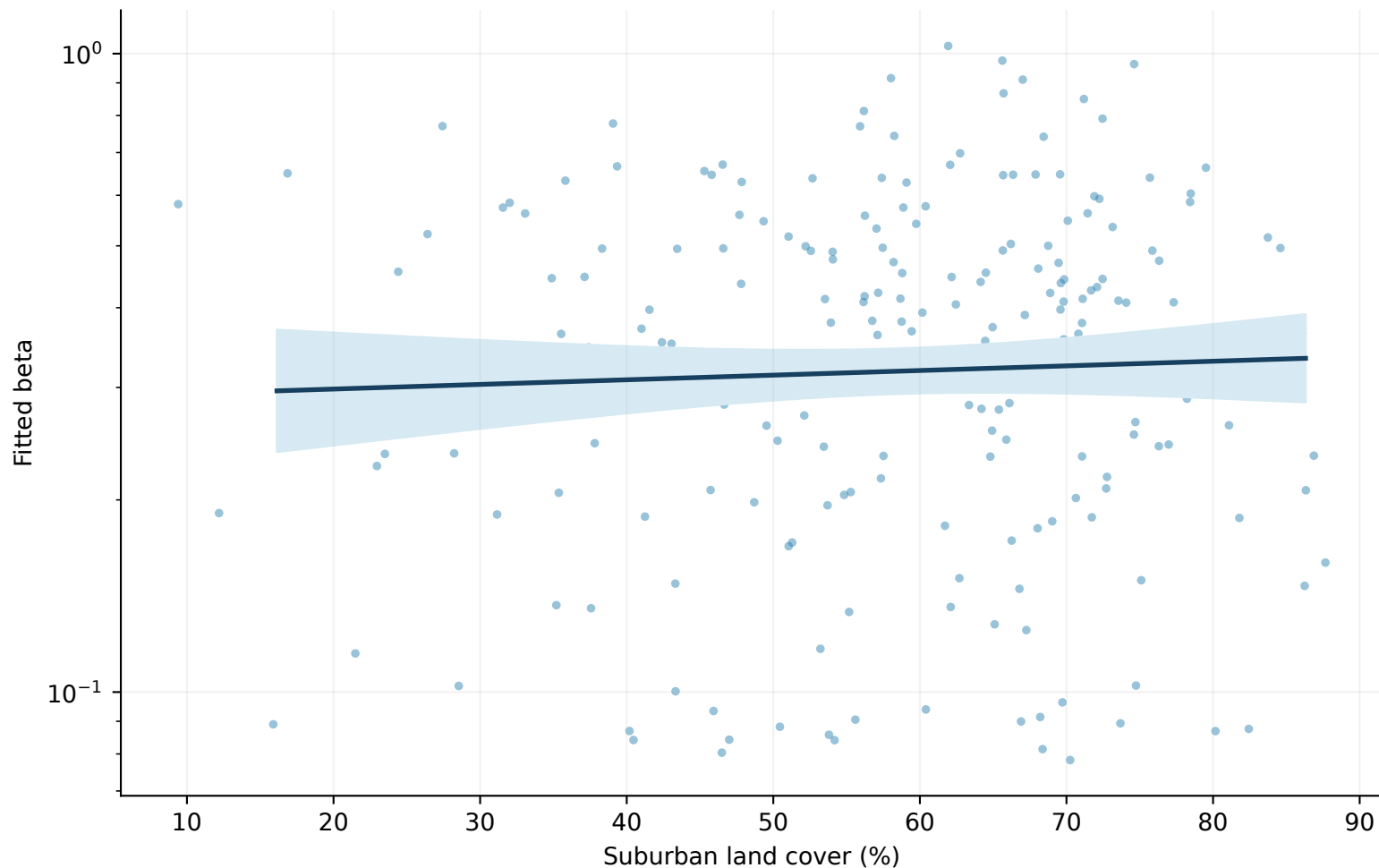
Fitted beta vs Urban land cover



n=227 | effect +23.1% per 10 percentage points | 95% CI [+14.5, +32.3]% | p=1.78e-08 | q=1.15e-07 | R²=0.106

A 10 percentage points increase in Urban land cover is associated with an estimated 23.1% higher fitted beta; unadjusted and non-causal.

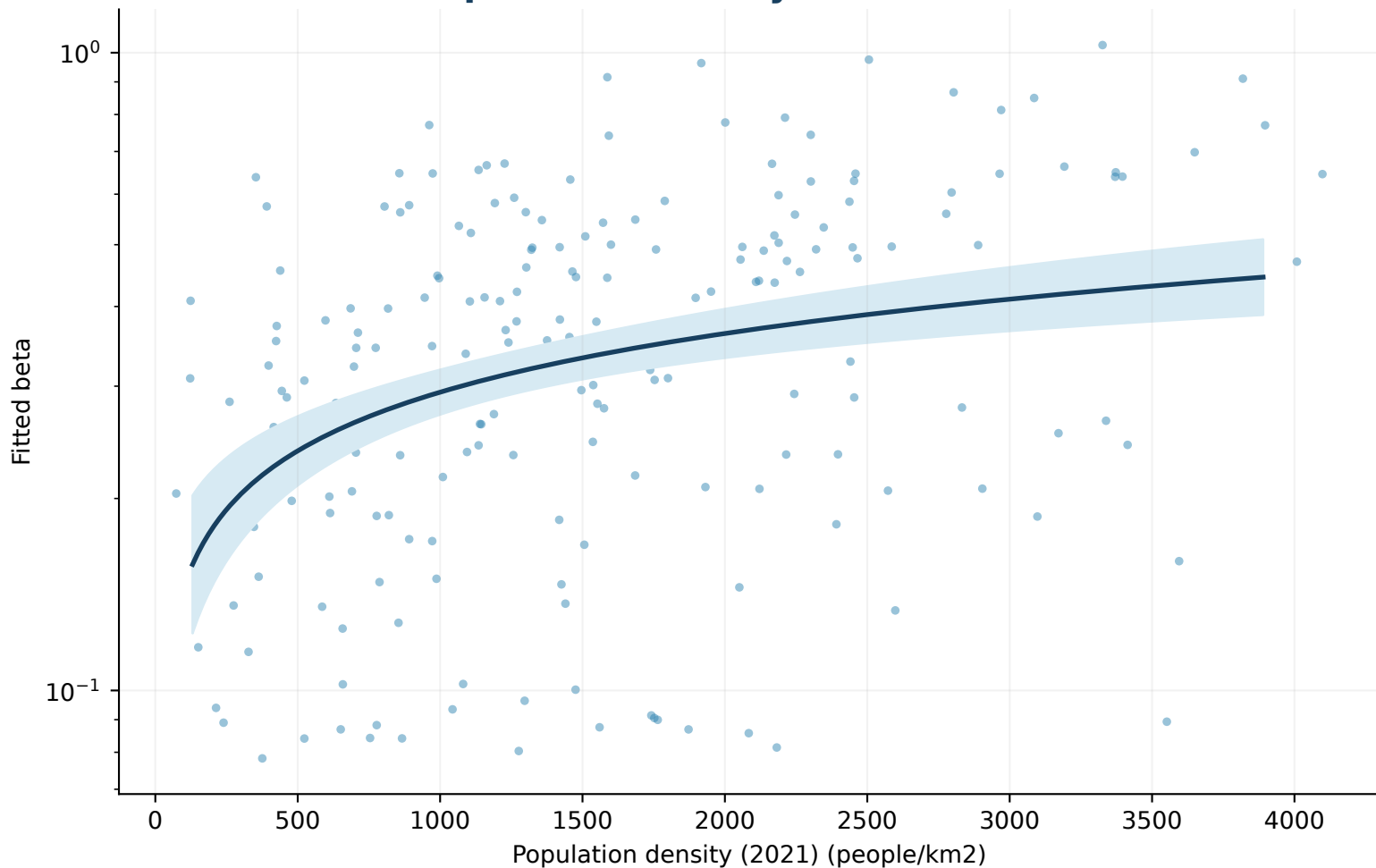
Fitted beta vs Suburban land cover



n=227 | effect +1.7% per 10 percentage points | 95% CI [-3.1, +6.7]% | p=0.499 | q=0.541 | R²=0.002

A 10 percentage points increase in Suburban land cover is associated with an estimated 1.7% higher fitted beta; unadjusted and non-causal.

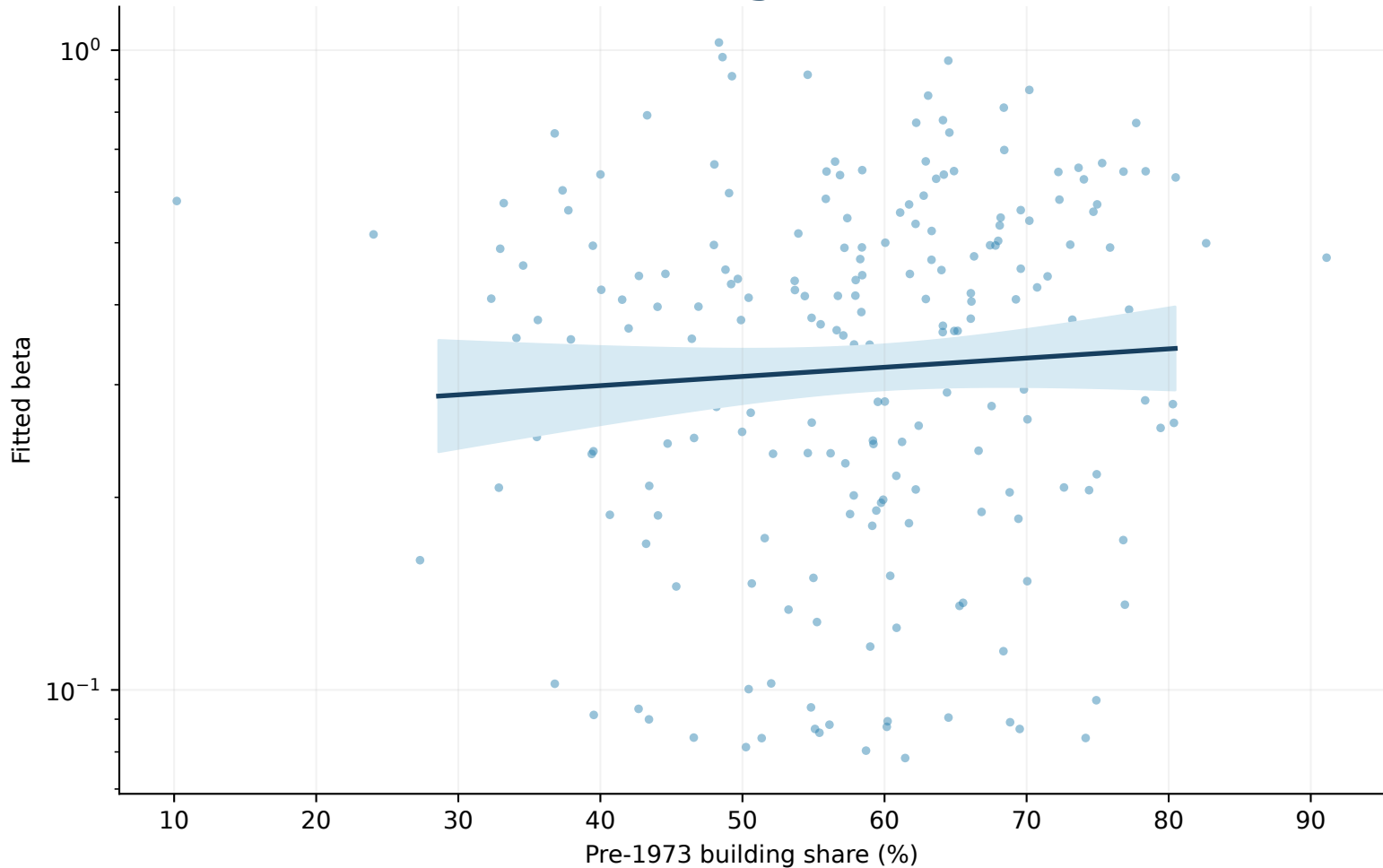
Fitted beta vs Population density (2021)



n=227 | effect +23.6% per doubling | 95% CI [+15.1, +32.8]% | p=6.26e-09 | q=8.14e-08 | R²=0.132

A doubling increase in Population density (2021) is associated with an estimated 23.6% higher fitted beta; unadjusted and non-causal.

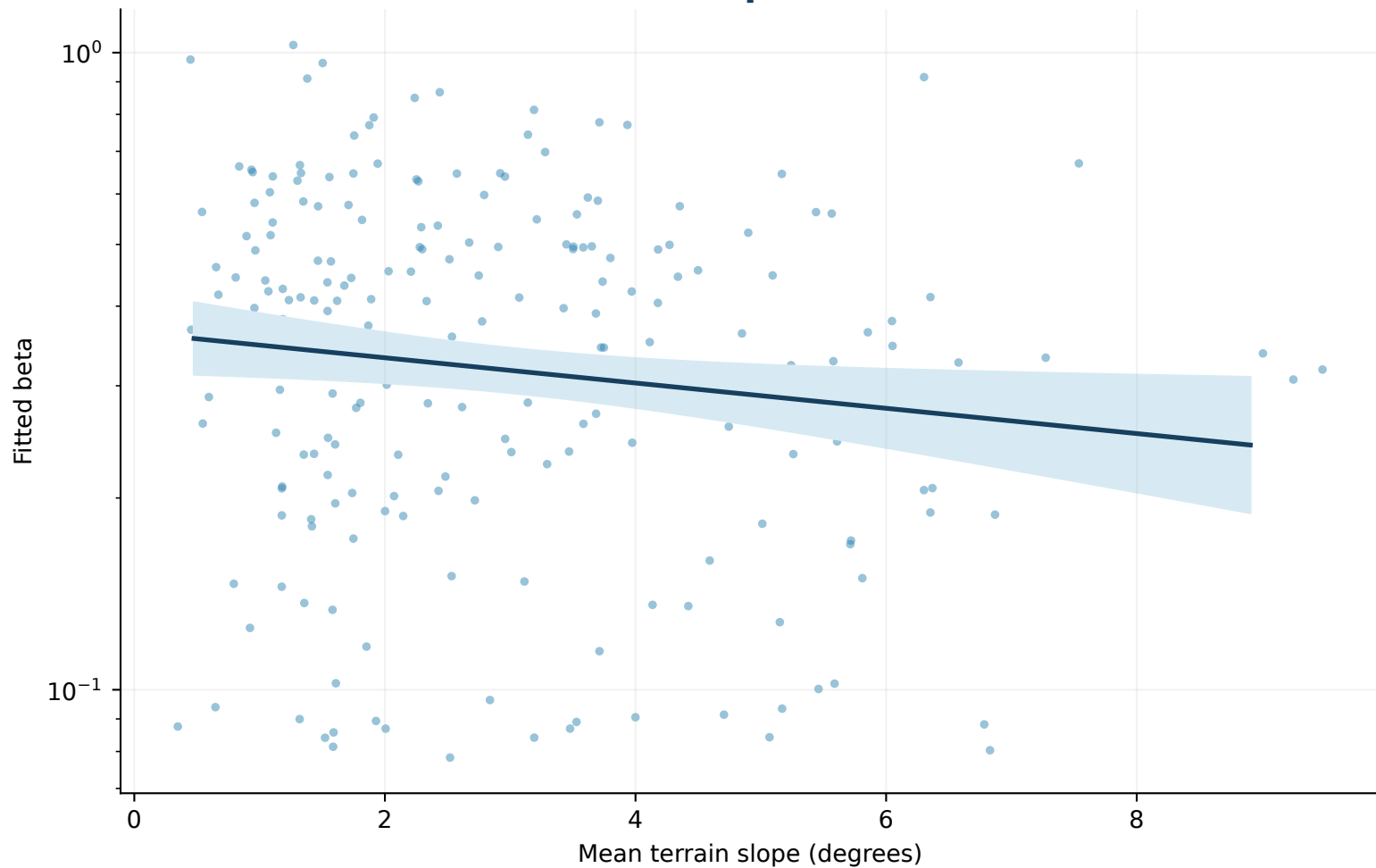
Fitted beta vs Pre-1973 building share



n=227 | effect +3.4% per 10 percentage points | 95% CI [-2.6, +9.7]% | p=0.278 | q=0.329 | R²=0.005

A 10 percentage points increase in Pre-1973 building share is associated with an estimated 3.4% higher fitted beta; unadjusted and non-causal.

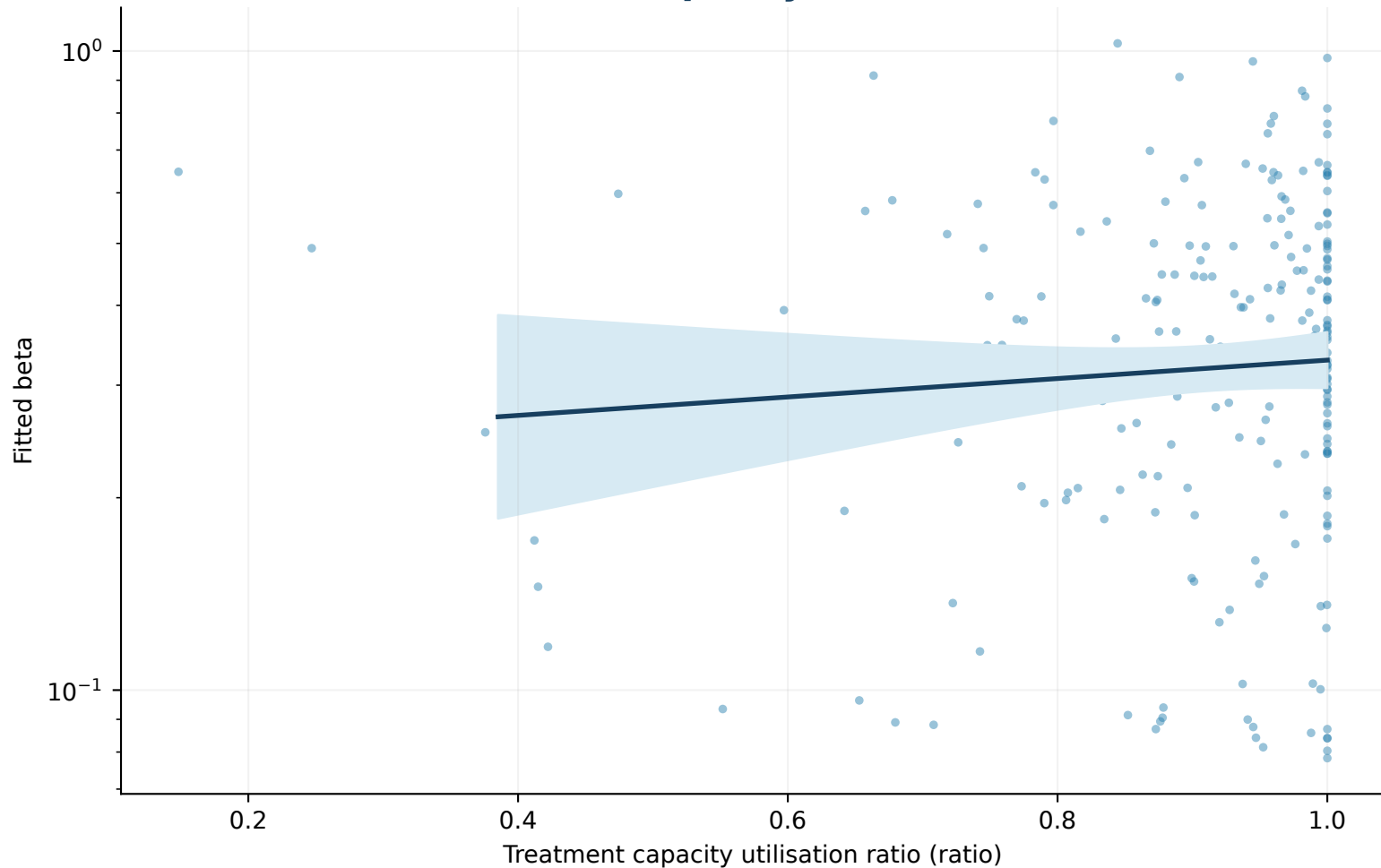
Fitted beta vs Mean terrain slope



n=227 | effect -4.5% per 1 degree | 95% CI [-8.2, -0.6]% | p=0.0238 | q=0.0338 | R2=0.020

A 1 degree increase in Mean terrain slope is associated with an estimated 4.5% lower fitted beta; unadjusted and non-causal.

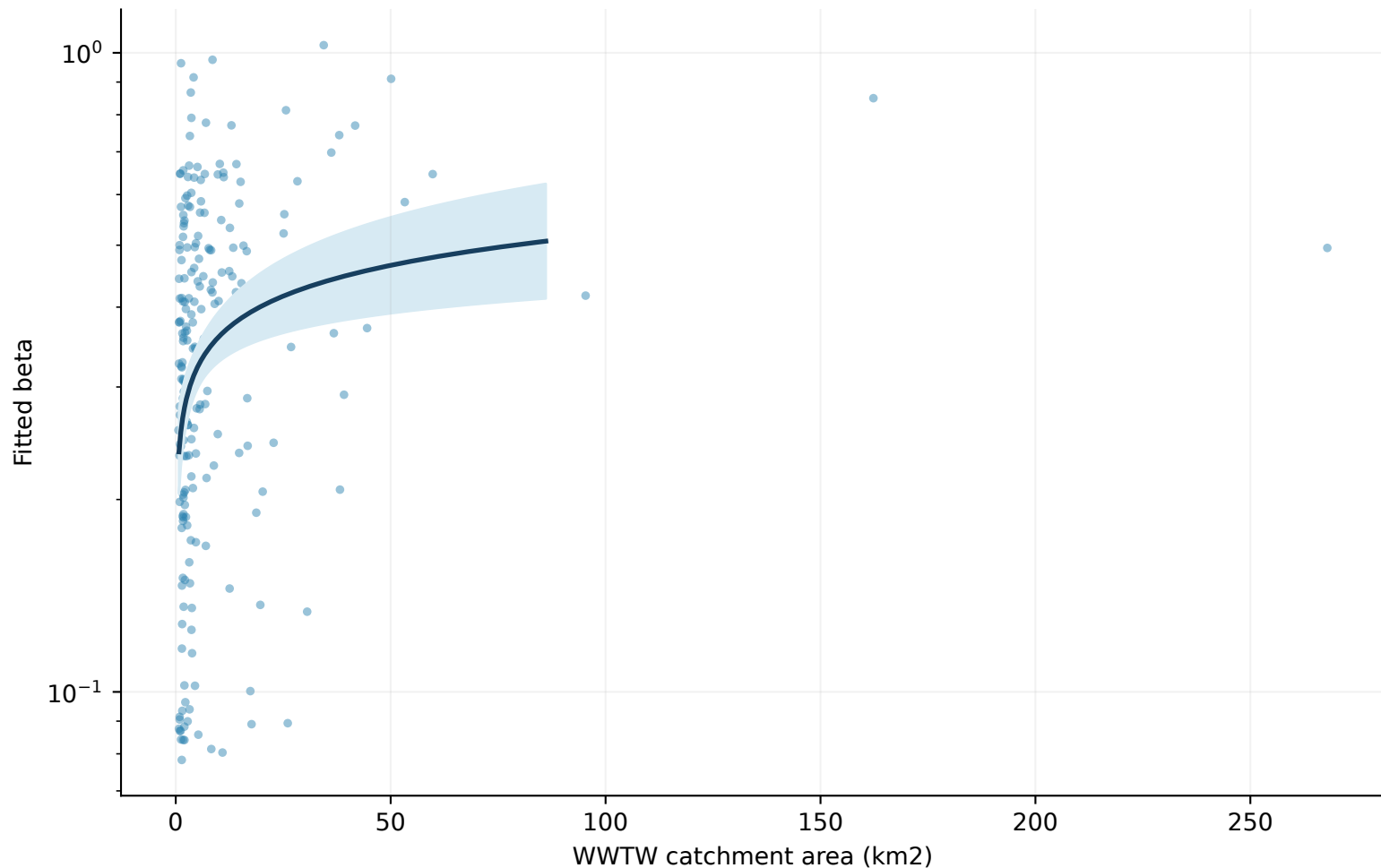
Fitted beta vs Treatment capacity utilisation ratio



n=227 | effect +3.4% per 0.1 ratio units | 95% CI [-3.4, +10.6]% | p=0.336 | q=0.336 | R2=0.006

A 0.1 ratio units increase in Treatment capacity utilisation ratio is associated with an estimated 3.4% higher fitted beta; unadjusted and non-causal

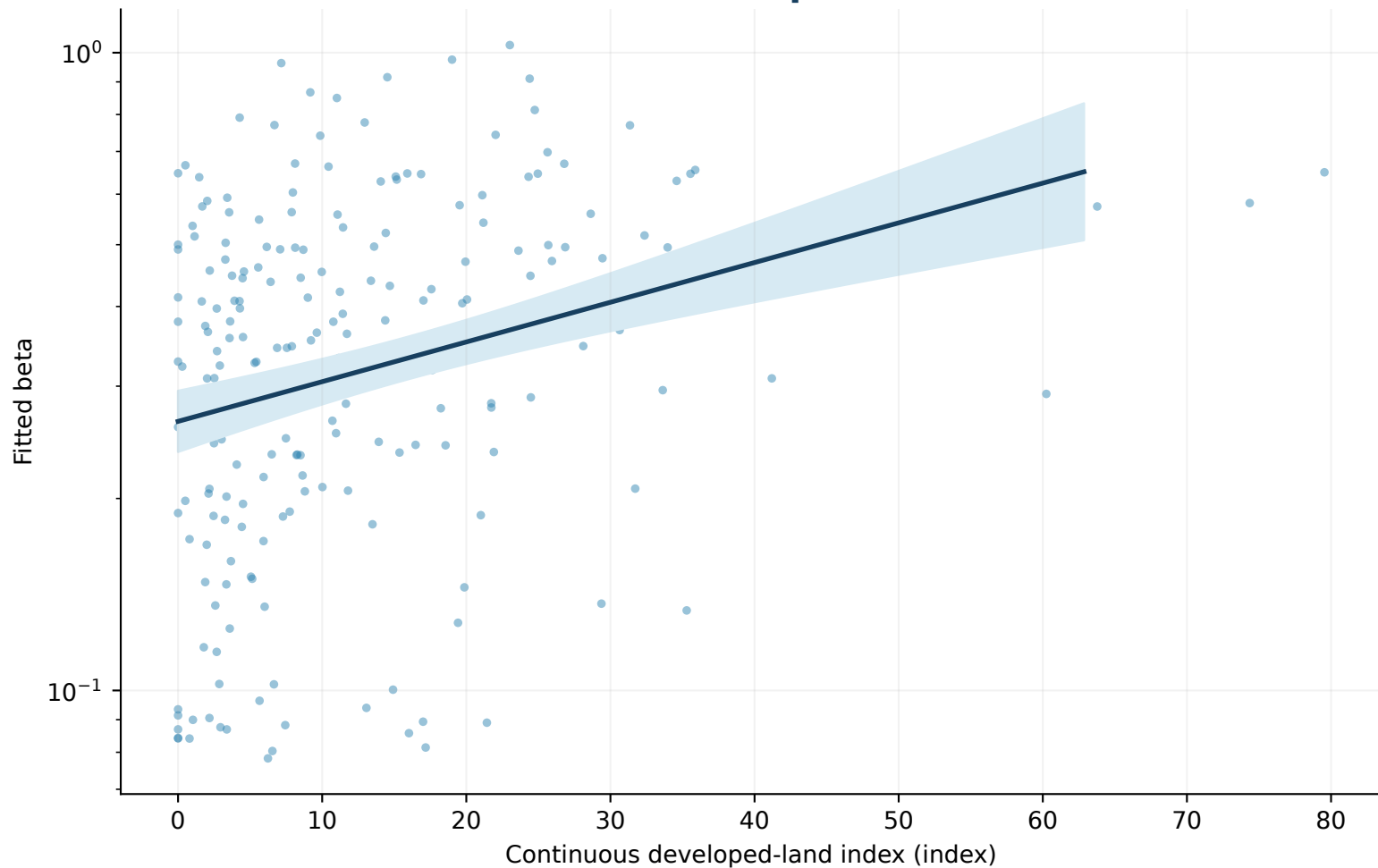
Fitted beta vs WWTW catchment area



n=227 | effect +11.7% per doubling | 95% CI [+6.6, +17.1]% | p=3.65e-06 | q=7.3e-06 | R²=0.086

A doubling increase in WWTW catchment area is associated with an estimated 11.7% higher fitted beta; unadjusted and non-causal.

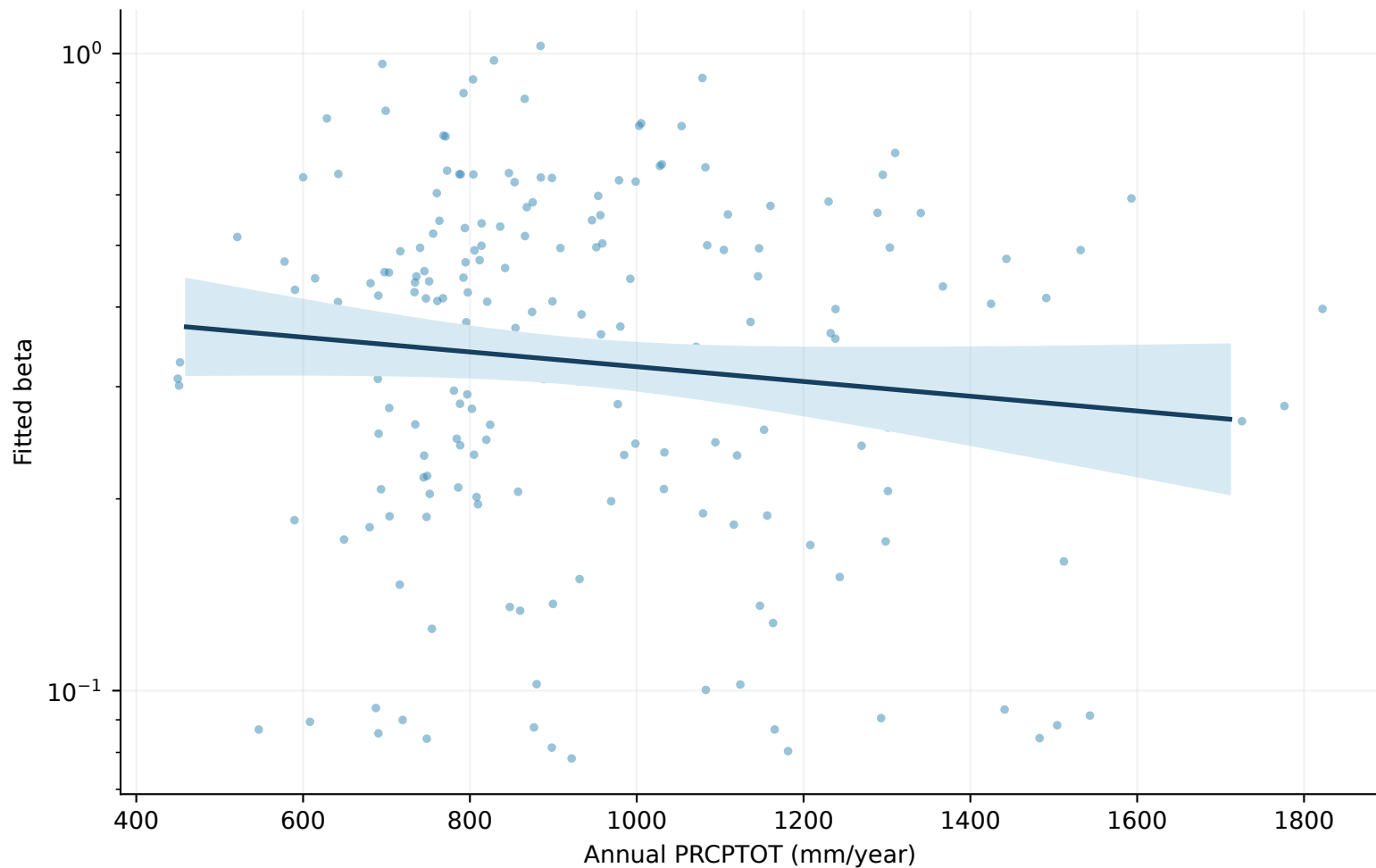
Fitted beta vs Continuous developed-land index



n=227 | effect +15.4% per 10 index points | 95% CI [+9.8, +21.3]% | p=1.8e-08 | q=1.8e-08 | R²=0.090

A 10 index points increase in Continuous developed-land index is associated with an estimated 15.4% higher fitted beta; unadjusted and non-cau

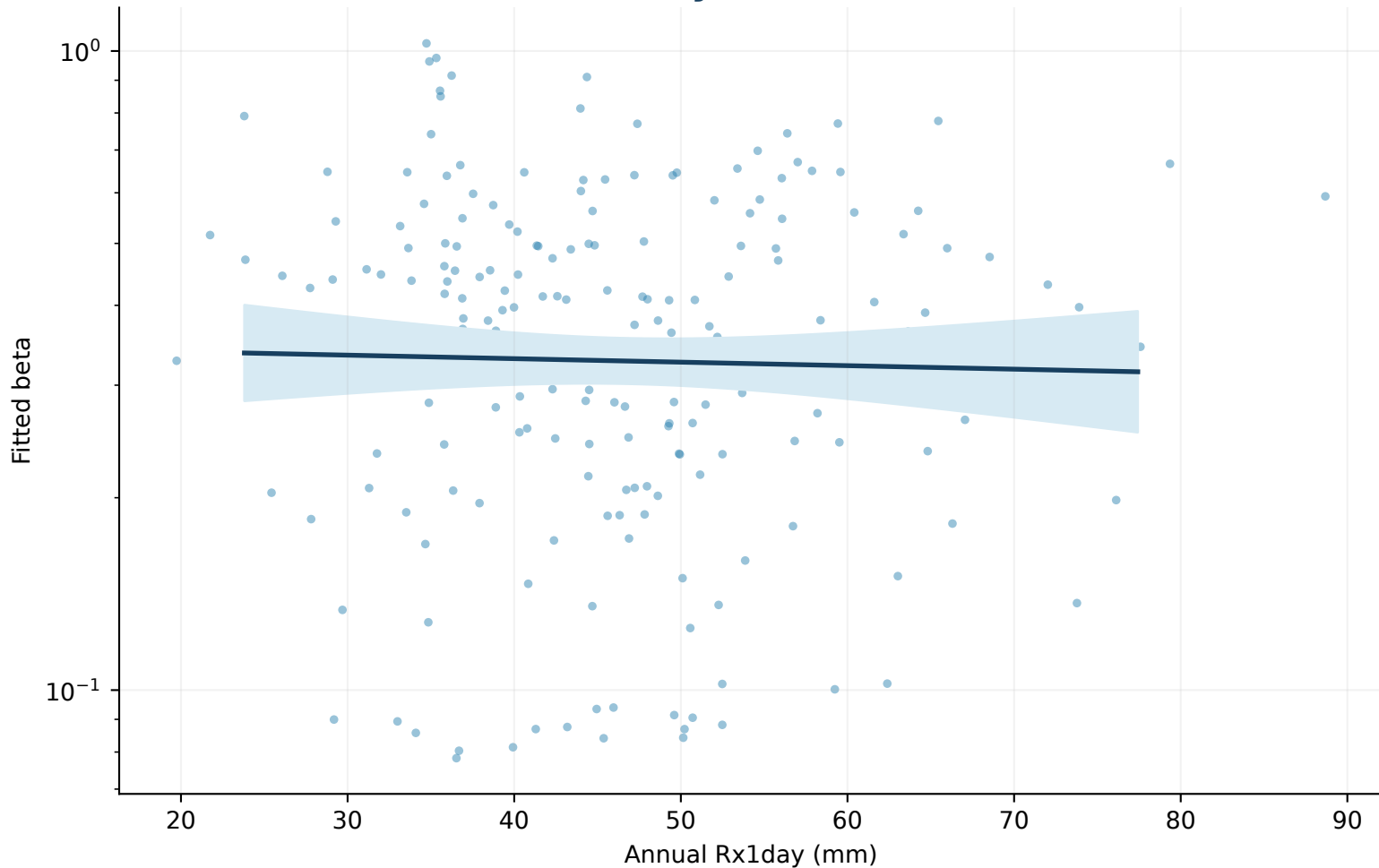
Fitted beta vs Annual PRCPTOT



n=212 | effect -2.6% per 100 mm/year | 95% CI [-5.8, +0.6]% | p=0.111 | q=0.185 | R2=0.013

A 100 mm/year increase in Annual PRCPTOT is associated with an estimated 2.6% lower fitted beta; unadjusted and non-causal.

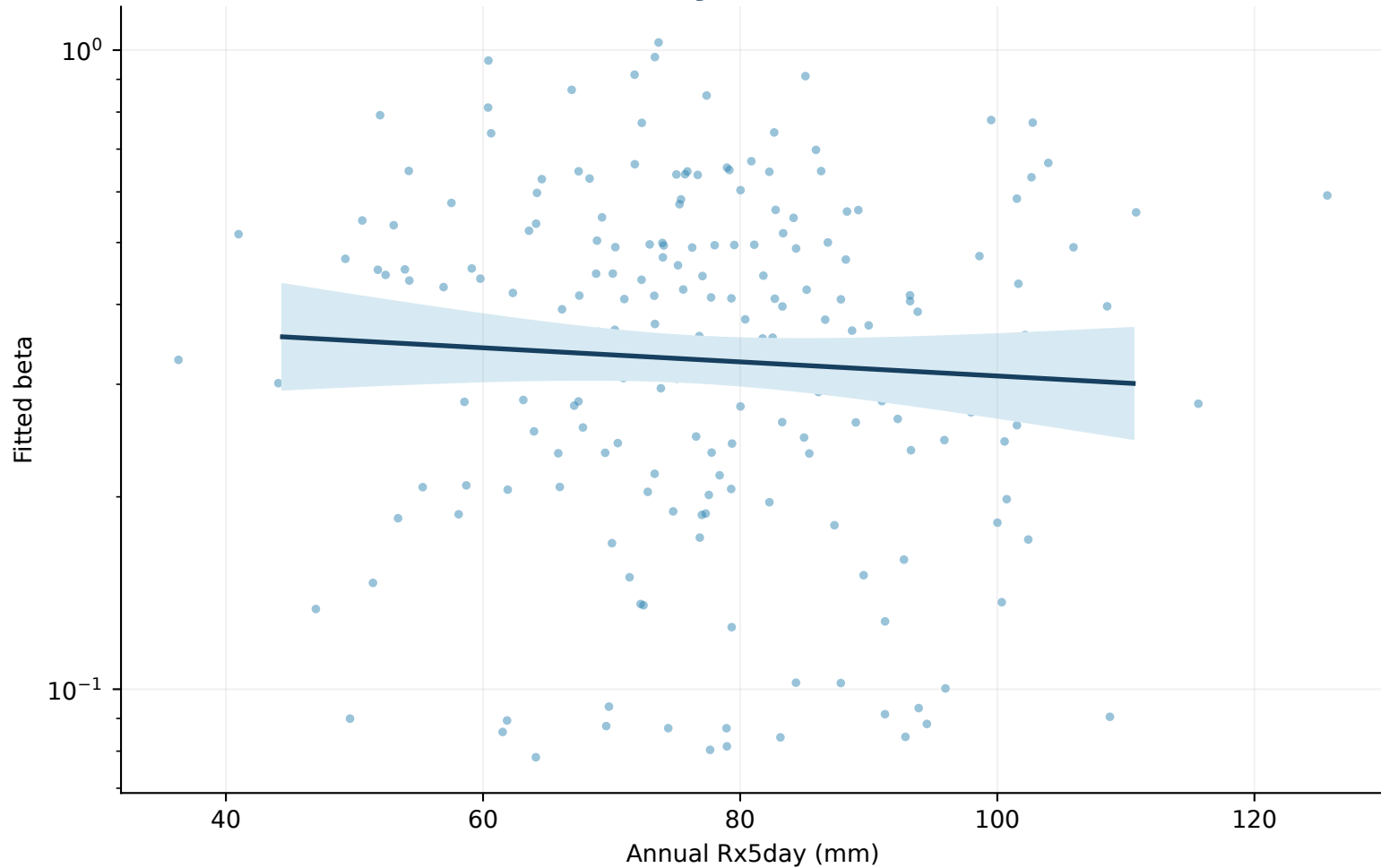
Fitted beta vs Annual Rx1day



n=212 | effect -1.3% per 10 mm | 95% CI [-7.6, +5.5]% | p=0.708 | q=0.708 | R²=0.001

A 10 mm increase in Annual Rx1day is associated with an estimated 1.3% lower fitted beta; unadjusted and non-causal.

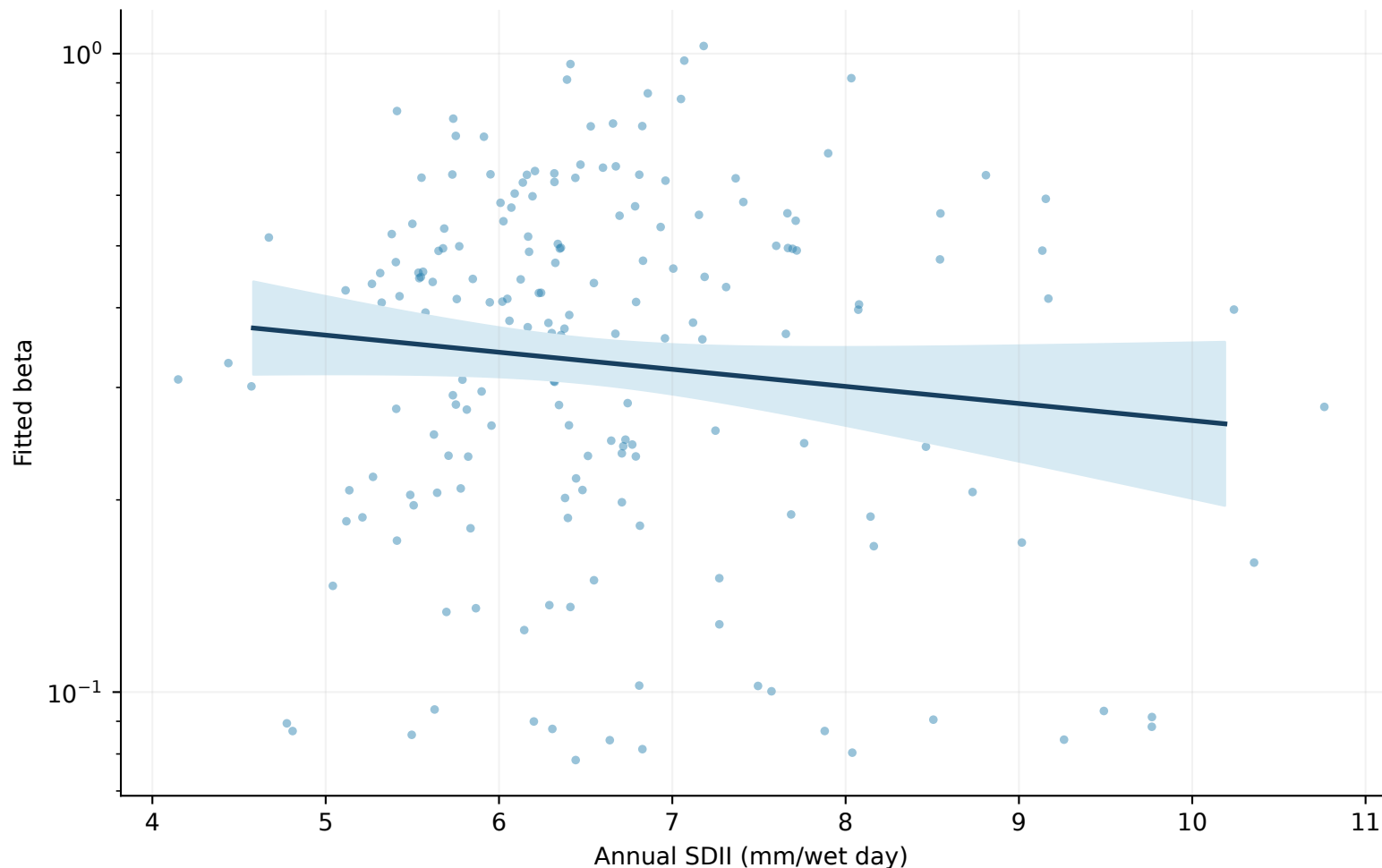
Fitted beta vs Annual Rx5day



n=212 | effect -6.1% per 25 mm | 95% CI [-17.8, +7.2]% | p=0.35 | q=0.4 | R²=0.004

A 25 mm increase in Annual Rx5day is associated with an estimated 6.1% lower fitted beta; unadjusted and non-causal.

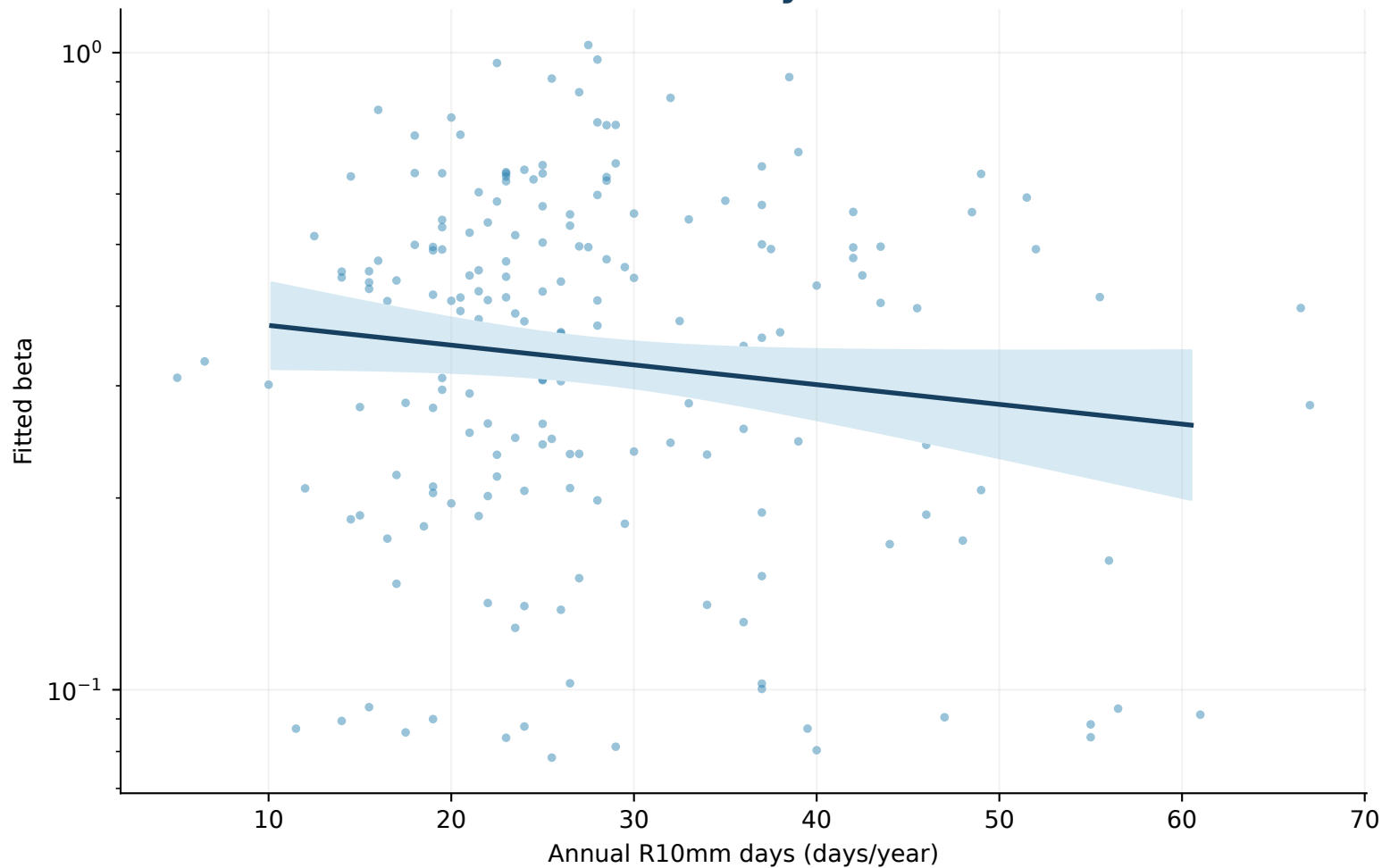
Fitted beta vs Annual SDII



n=212 | effect -6.0% per 1 mm/wet day | 95% CI [-12.9, +1.5]% | p=0.115 | q=0.185 | R2=0.014

A 1 mm/wet day increase in Annual SDII is associated with an estimated 6.0% lower fitted beta; unadjusted and non-causal.

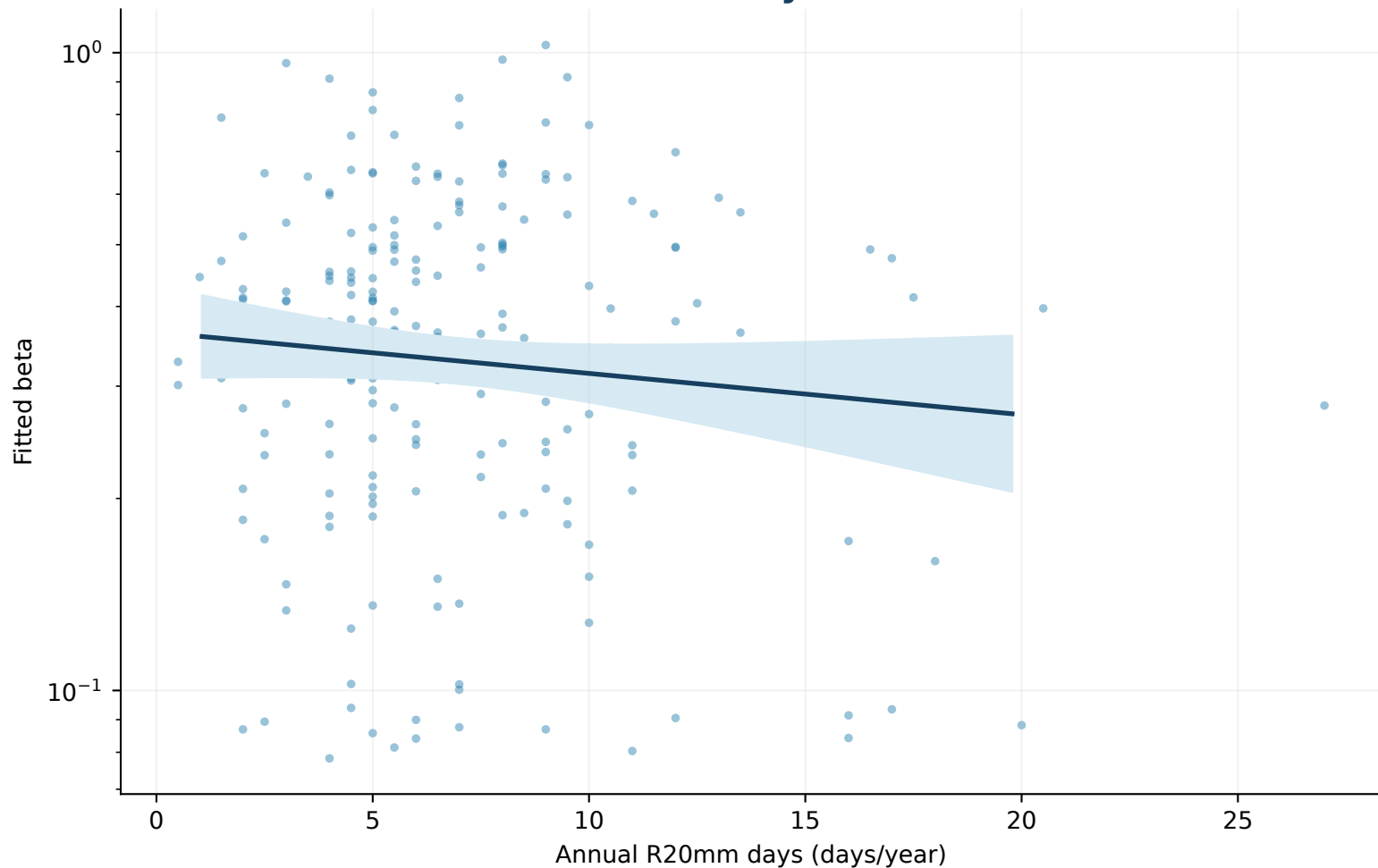
Fitted beta vs Annual R10mm days



n=212 | effect -3.5% per 5 days/year | 95% CI [-7.2, +0.3]% | p=0.0712 | q=0.185 | R²=0.018

A 5 days/year increase in Annual R10mm days is associated with an estimated 3.5% lower fitted beta; unadjusted and non-causal.

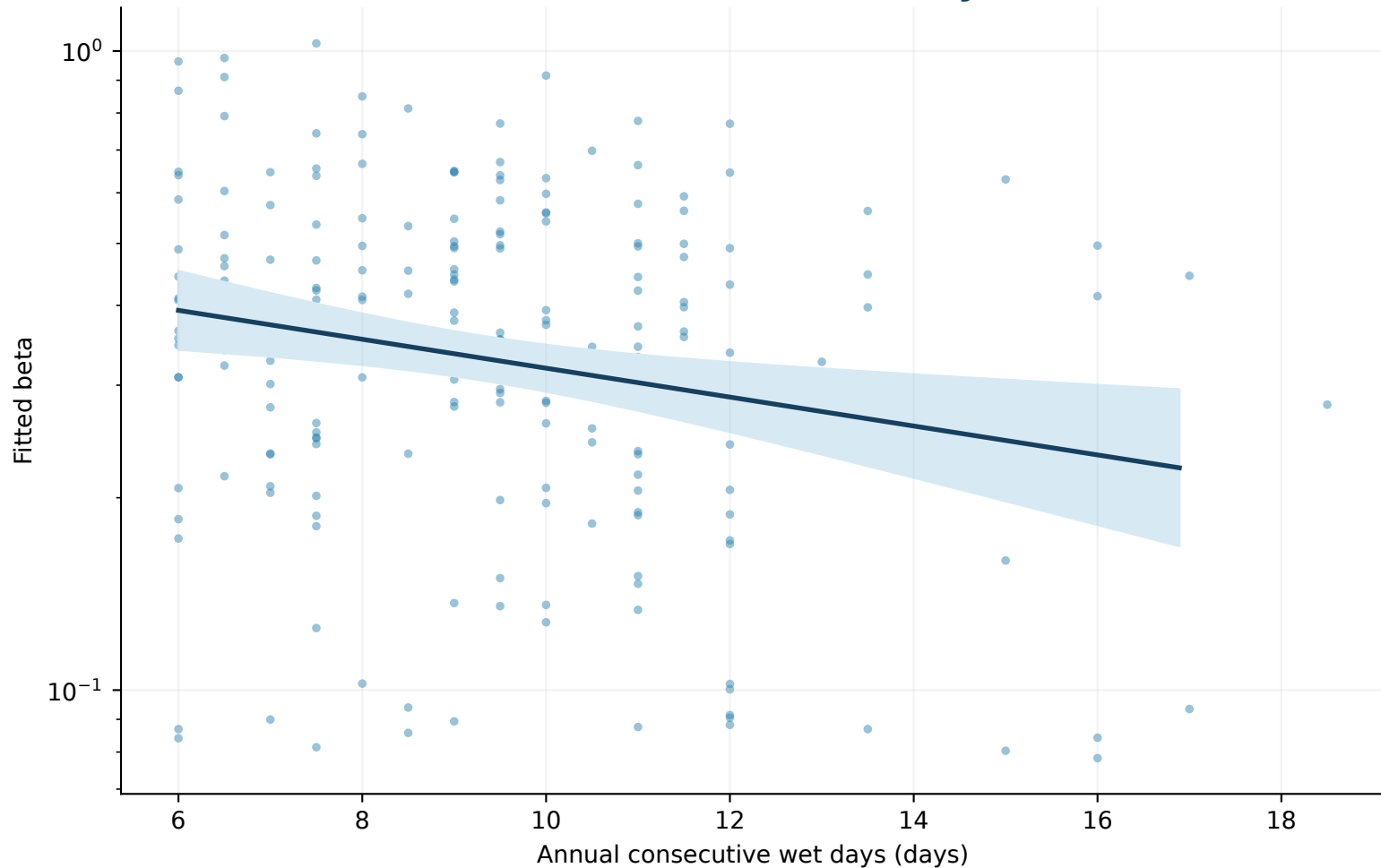
Fitted beta vs Annual R20mm days



n=212 | effect -7.2% per 5 days/year | 95% CI [-16.4, +3.1]% | p=0.164 | q=0.219 | R²=0.010

A 5 days/year increase in Annual R20mm days is associated with an estimated 7.2% lower fitted beta; unadjusted and non-causal.

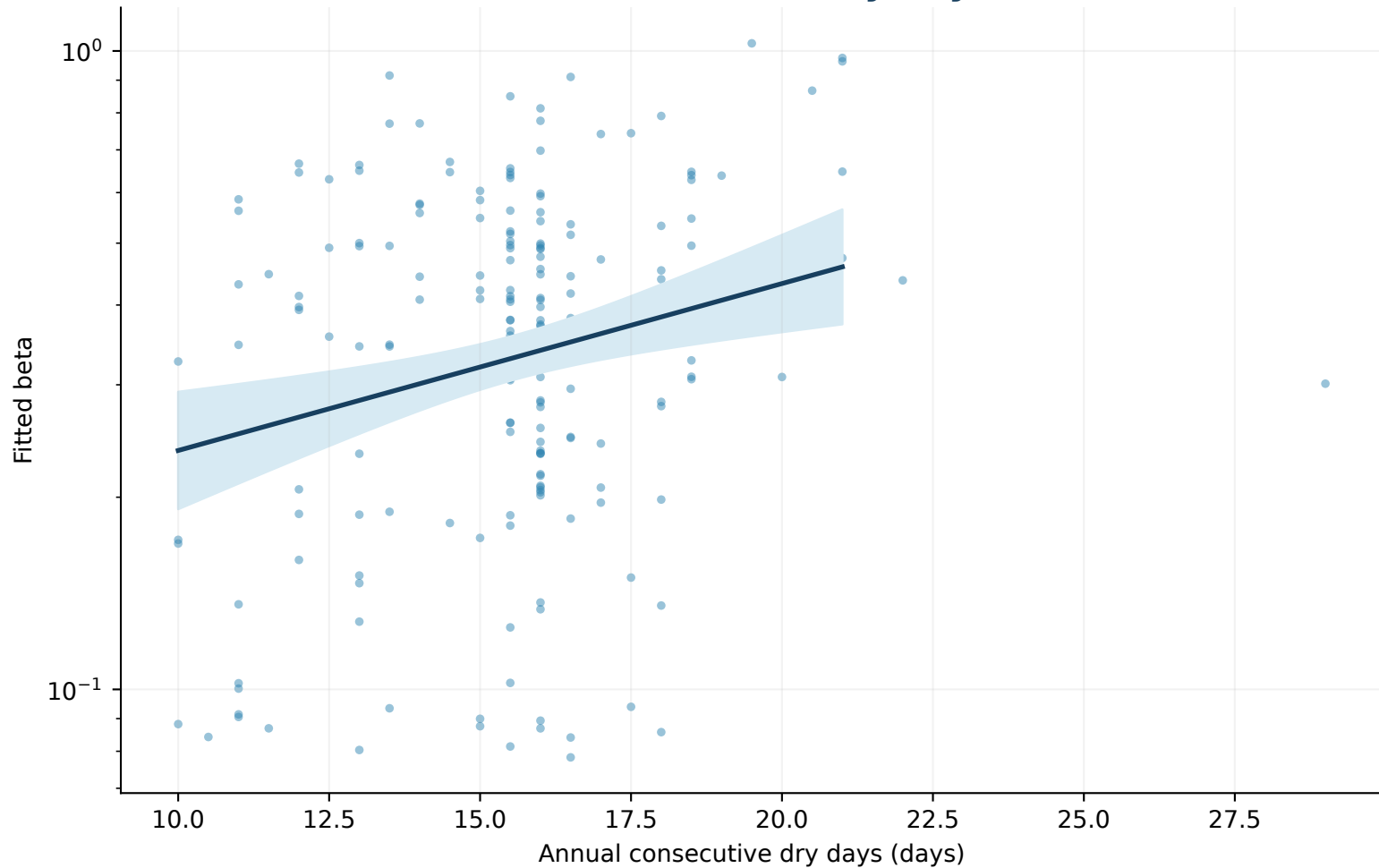
Fitted beta vs Annual consecutive wet days



n=212 | effect -5.1% per 1 day | 95% CI [-8.4, -1.6]% | p=0.0041 | q=0.0164 | R²=0.048

A 1 day increase in Annual consecutive wet days is associated with an estimated 5.1% lower fitted beta; unadjusted and non-causal.

Fitted beta vs Annual consecutive dry days



n=212 | effect +35.1% per 5 days | 95% CI [+13.3, +61.1]% | p=0.000793 | q=0.00635 | R²=0.066

A 5 days increase in Annual consecutive dry days is associated with an estimated 35.1% higher fitted beta; unadjusted and non-causal.

Results summary

Continuous developed-land index: +15.4% [+9.8, +21.3]%; $p=1.8e-08$; $q=1.8e-08$; FDR-supported.

Population density (2021): +23.6% [+15.1, +32.8]%; $p=6.26e-09$; $q=8.14e-08$; FDR-supported.

Urban land cover: +23.1% [+14.5, +32.3]%; $p=1.78e-08$; $q=1.15e-07$; FDR-supported.

WWTW catchment area: +11.7% [+6.6, +17.1]%; $p=3.65e-06$; $q=7.3e-06$; FDR-supported.

Low-productivity hydrogeology: -4.5% [-6.5, -2.5]%; $p=1.18e-05$; $q=5.12e-05$; FDR-supported.

Fracture-flow hydrogeology: -3.0% [-4.8, -1.2]%; $p=0.00144$; $q=0.00468$; FDR-supported.

Annual consecutive dry days: +35.1% [+13.3, +61.1]%; $p=0.000793$; $q=0.00635$; FDR-supported.

Static winter rainfall (1991-2020): -12.3% [-19.7, -4.3]%; $p=0.00342$; $q=0.00888$; FDR-supported.

Static annual rainfall (1991-2020): -4.8% [-8.1, -1.5]%; $p=0.00494$; $q=0.0107$; FDR-supported.

Intergranular-flow hydrogeology: +4.1% [+1.0, +7.2]%; $p=0.00834$; $q=0.0155$; FDR-supported.

Annual consecutive wet days: -5.1% [-8.4, -1.6]%; $p=0.0041$; $q=0.0164$; FDR-supported.

Moderate-productivity hydrogeology: +2.2% [+0.5, +4.0]%; $p=0.0128$; $q=0.0207$; FDR-supported.